How-To Library

Informatica

Importing Metadata From a Netezza Connection in Test Data Management

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Abstract

You can use data in a Netezza database to perform tasks in Test Data Management (TDM).

This article describes the steps to create a Netezza connection and import metadata from a Netezza connection in Test Data Manager. This article assumes that you have knowledge of TDM and are familiar with TDM operations.

Supported Versions

- Test Data Management 10.2.0
- Test Data Management 10.1.0
- Test Data Management 9.7.1 HotFix 1
- Test Data Management 9.7.1

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Overview

To perform TDM operations on source data, you must import the source metadata into the TDM repository.

You can perform data subset, data masking, and data generation operations on data from a Netezza database. Create a 32-bit ODBC connection to import source metadata into PowerCenter, and then create connections to perform TDM operations on the data in Test Data Manager. When you create a project, add one or more sources to the project. You can then create constraints to create relationships between the sources and apply filter criteria to perform data subset operations. You can apply data masking or data generation rules to perform data masking or data generation.

Restrictions

The following restrictions apply to TDM operations that use Netezza connections:

- You cannot import metadata from a Netezza data source directly into Test Data Manager. You must import the metadata from PowerCenter.
- You cannot test the connection when you create a Netezza connection in Test Data Manager.
- You cannot disable and enable indexes and constraints in a Netezza connection in Test Data Manager. The property appears in a plan that uses Netezza connections, but does not work as expected.

Example Scenario

You work with a software product testing team that creates and manages its test data in TDM.

You need to create test data from a Netezza data source. To use data from a Netezza database, you import the metadata into Test Data Manager. You can then run a TDM operation to create the test data.

To import metadata from a Netezza data source into Test Data Manager, perform the following tasks:

1. Configure a 32-bit data source name (DSN) with 32-bit Netezza ODBC drivers on the PowerCenter Client machine.
2. Configure a 64-bit DSN with 64-bit Netezza drivers on the Informatica server machine.
3. Import the source metadata into PowerCenter using the 32-bit DSN created in step 1.
4. Create a project in Test Data Manager and import the source metadata from the PowerCenter repository.
5. Create a Netezza connection in Test Data Manager. You use the connection in a plan to perform TDM operations.

Configure a 32-bit Netezza ODBC DSN

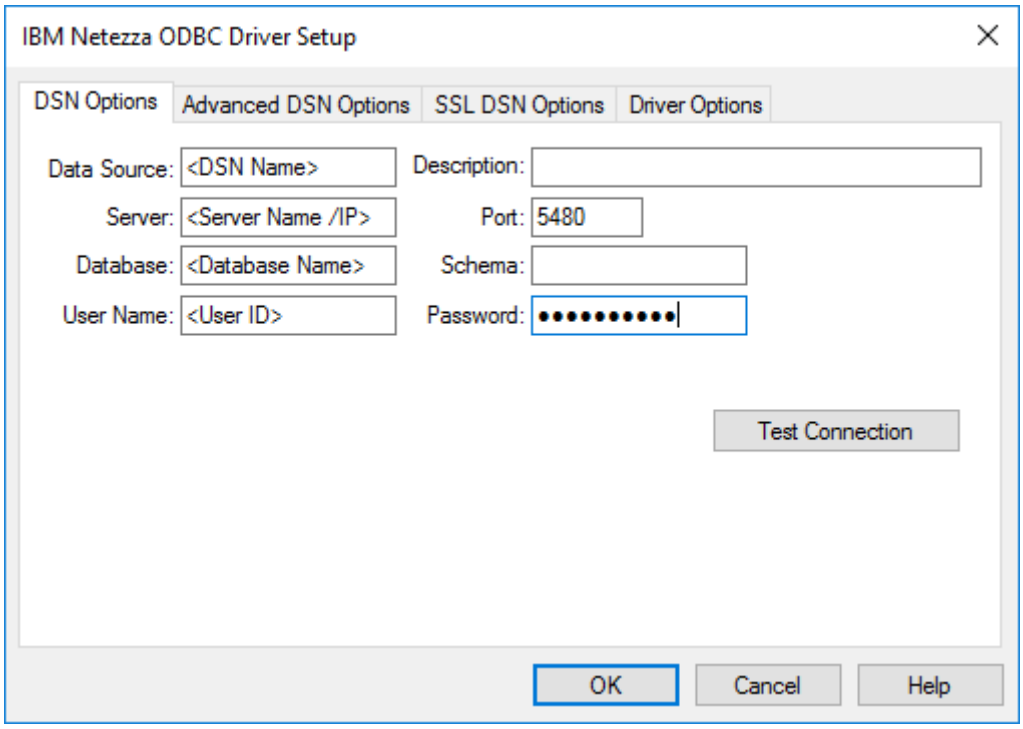
PowerCenter client requires a 32-bit Netezza ODBC DSN to import metadata from a Netezza database. To import metadata into PowerCenter, configure a 32-bit Netezza ODBC data source that connects to a Netezza server on the PowerCenter Client machine.

Install Netezza 32-bit client before you configure the DSN.

1. On the PowerCenter Client machine, open ODBC Data Source Administrator (32-bit).
2. Click **System DSN** > **Add**.

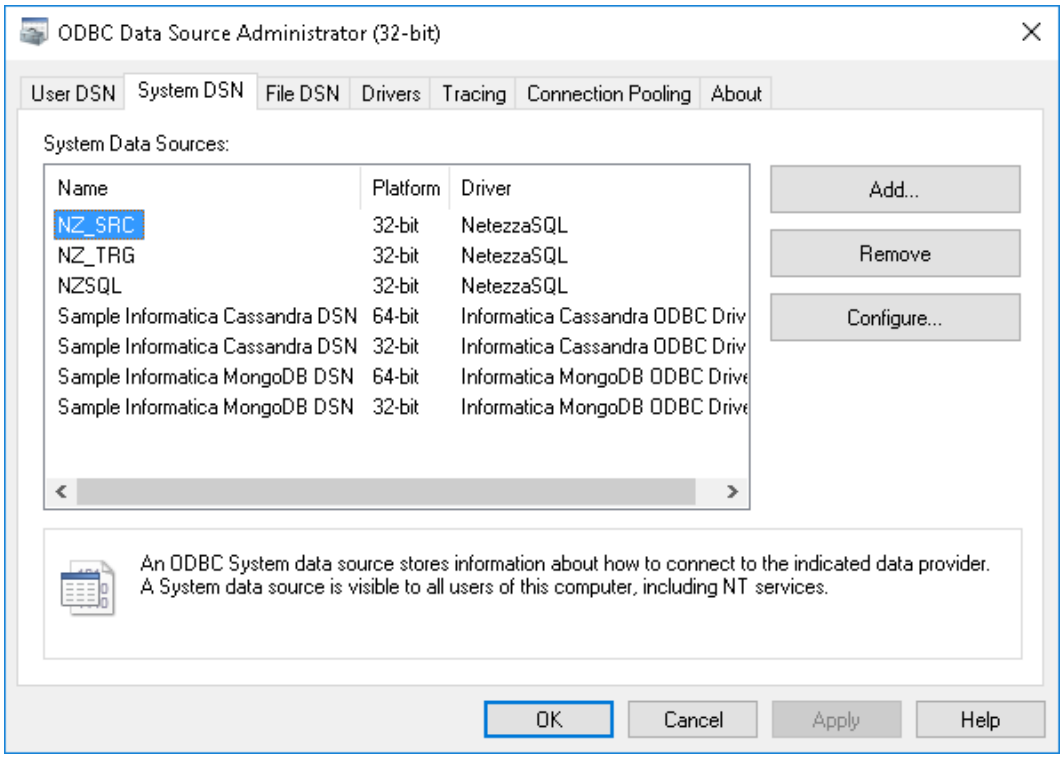
   The **Create New Data Source** window opens.
3. Select the NetezzaSQL driver and click **Finish**.
4. In the **IBM Netezza ODBC Driver Setup** window, enter a data source name and description. Enter the server, port, and other database details.

The following image shows the **IBM Netezza ODBC Driver Setup** window:



5. Test the connection to validate the information.

   The data source is created and appears in the list of system data sources.
   The following image shows the list of data sources in the **ODBC Data Source Administrator** window:



6. Click **OK**.

Configure a 64-bit Netezza ODBC DSN on the Informatica Server

Informatica requires a 64-bit data source to perform data movement. Configure a 64-bit Netezza ODBC data source for each Netezza database that you want to access to perform TDM operations. You can create a 64-bit Netezza ODBC data source on Windows and Linux machines. This task describes the steps to configure the data source name on a Linux machine. The steps differ for Windows machines.

Install the Linux Netezza 64-bit client on the Informatica Server machine.

1. On the Informatica server, open the `<Informatica install directory>\ODBC7.1\odbcinst.ini` file.
2. Configure the Netezza ODBC driver details. Add an entry with the following properties:

```
[Netezza SQL]
Driver = <Netezza client install folder>/linux64/lib/libnzsqlodbc3.so
Setup = <Netezza client install folder>Netezza/linux64/lib/libnzsqlodbc3.so
APILevel = 1
ConnectFunctions = YYN
Description = Netezza ODBC driver
DriverODBCVer    = 03.51
DebugLogging     = false
LogPath          = /tmp
UnicodeTranslationOption = utf8
CharacterTranslationOption = all
PreFetch         = 256
Socket           = 16384
```

3. Save and close the file.
4. Open the `<Informatica install directory>\ODBC7.1\odbc.ini` file.
5. Add an entry for each data source that you create in the [ODBC Data Sources] section. For example, add the following in the [ODBC Data Sources] section:

```
NZSQL_SRC=NetezzaSQL
```

6. Configure the data source properties to connect to the Netezza database. You use the connections to perform TDM operations. Add an entry with the following properties for each Netezza connection that you create:

```
[NZSQL_SRC]
Driver= <Netezza client install folder>/linux64/lib64/libnzodbc.so
Description= Netezza SQL
ODBCServername= <server name>
Port=5480
Database=<database name>
Username=<user name>
Password=<password>
ReadOnly=false
FastSelect=false
ShowSystemTables=false
LegacySQLTables=false
LoginTimeout=0
QueryTimeout=0
DateFormat=1
NumericAsChar=false
SQLBitOneZero=false
StripCRLF=false
securityLevel=preferredUnSecured
caCertFile=
```

7. Save and close the file.
8. Set the following environment variables under .bash_profile.
   - `export NZ_ODBC_INI_PATH=$INFA_HOME/ODBC7.1`
   - `export ODBCINI=$ODBCHOME/odbc.ini`
9. Add the following entry to LD_LIBRARY_PATH. Maintain the order of the entries.

```
:$ODBCHOME/lib:/Home/Netezza/linux64/lib64
```

10. Optional. Run the following command to test the connection from `<Informatica install directory>/tools/debugtools/ssgodbc/linux64`.

    ```
    ./ssgodbc.linux64 -d <ODBC DSN> -u <user name> -p <password> -v
    ```

    The following output appears if the connection is valid:

    ```
    Connected
    ODBC version        = -03.52.0000-
    DBMS name           = -NetezzaSQL-
    DBMS version        = -06.00.0005 Release 6.0.5 (P-5) [Build 21415]-
    Driver name         = -libnzodbc.so-
    Driver version      = -Release 7.2.0.4-P3 [Build 44291]-
    Driver ODBC version = -03.51-

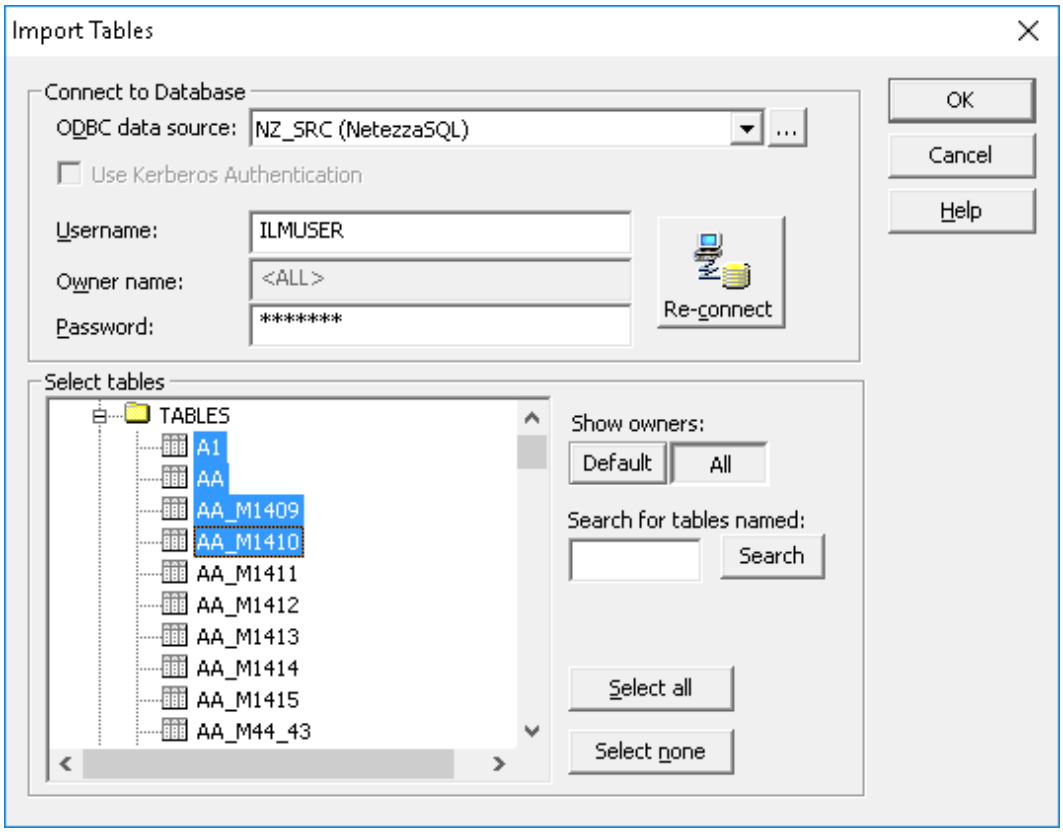
    Enter SQL string:
    ```

Import the Metadata into PowerCenter

Import source metadata into PowerCenter from Designer.

1. Open PowerCenter Designer and connect to the required repository.
2. Select the required folder.
3. Click **Sources** > **Import from Database**.

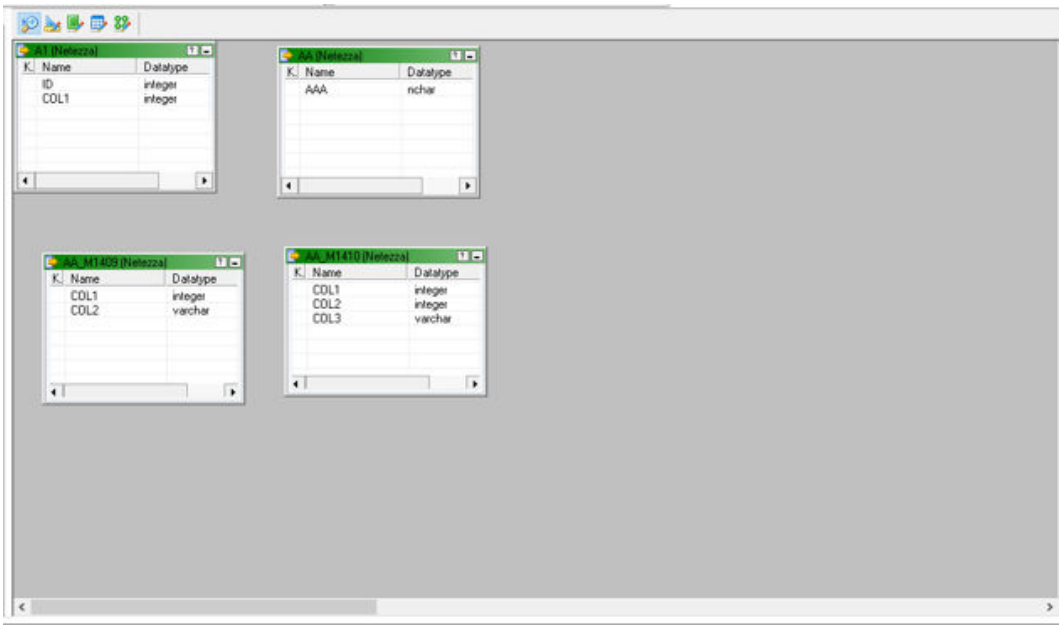
   The **Import Tables** window appears.
   The following image shows the **Import Tables** window:



4. Select the Netezza ODBC data source that you created on the PowerCenter Client machine.
5. Enter the user name, owner name, and password and click **Connect**.
6. Choose to show all owners.

7. Select the tables that you want to import and click **OK**.

   The tables are imported into the PowerCenter folder and open in the **Source Analyzer** window. The following image shows the **Source Analyzer** window in Designer:



8. Save the source metadata that you imported.

Create a Project and Import the Metadata in Test Data Manager

Create a project and import the metadata into the project to perform TDM operations on source data.

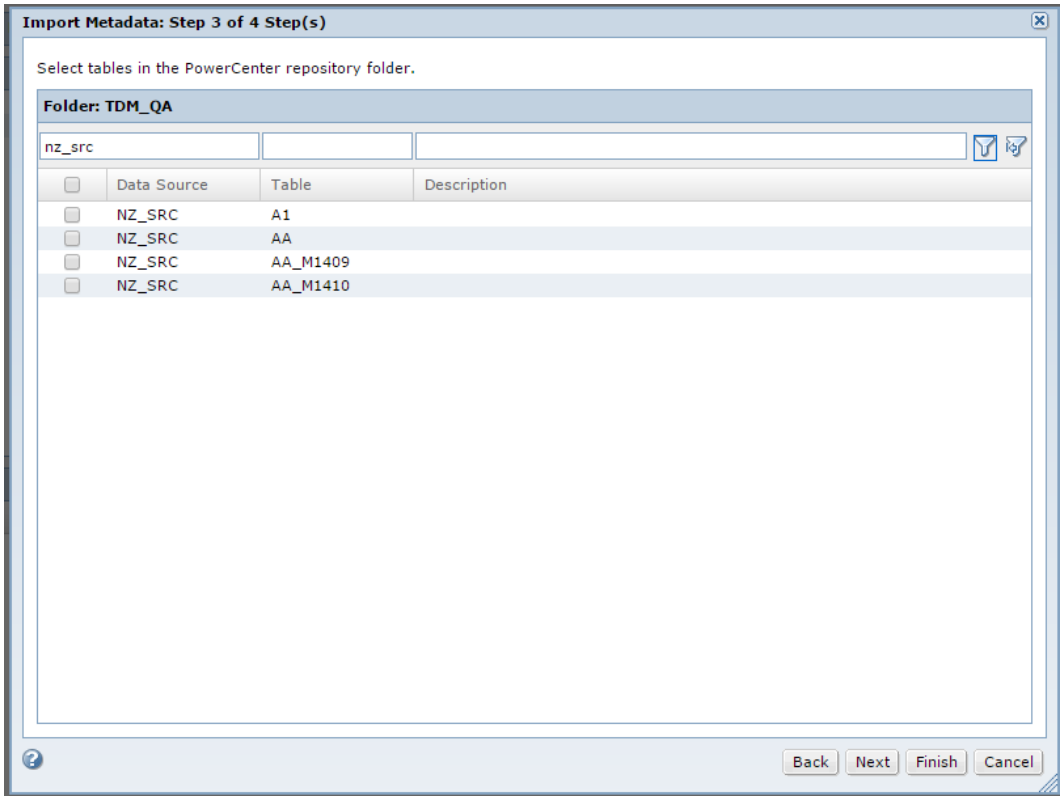
1. Log in to Test Data Manager.
2. Click **Projects** to access the projects.
3. Click **Actions** > **New**.

   The **Create Project** dialog box opens.
4. Enter a name and optional description and click **OK** to create the project.
5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.
6. Choose to import metadata from the PowerCenter repository and click **Next**.

   You cannot directly import metadata from a Netezza connection in Test Data Manager.
7. Select the folder into which you imported the source metadata in Designer and click **Next**.
8. Filter and select the required tables and click **Next**.

The following image shows the third step in the **Import Metadata** wizard:



9. Choose to import the metadata now and click **Finish**.

You can use the source metadata from the Netezza database to perform data subset or data masking operations in TDM.

Create a Netezza Connection in Test Data Manager

Create a Netezza connection in Test Data Manager to use the connection in a plan.

1. In the **Administrator** | **Connections** view, select **New Connection** from the **Actions** menu.

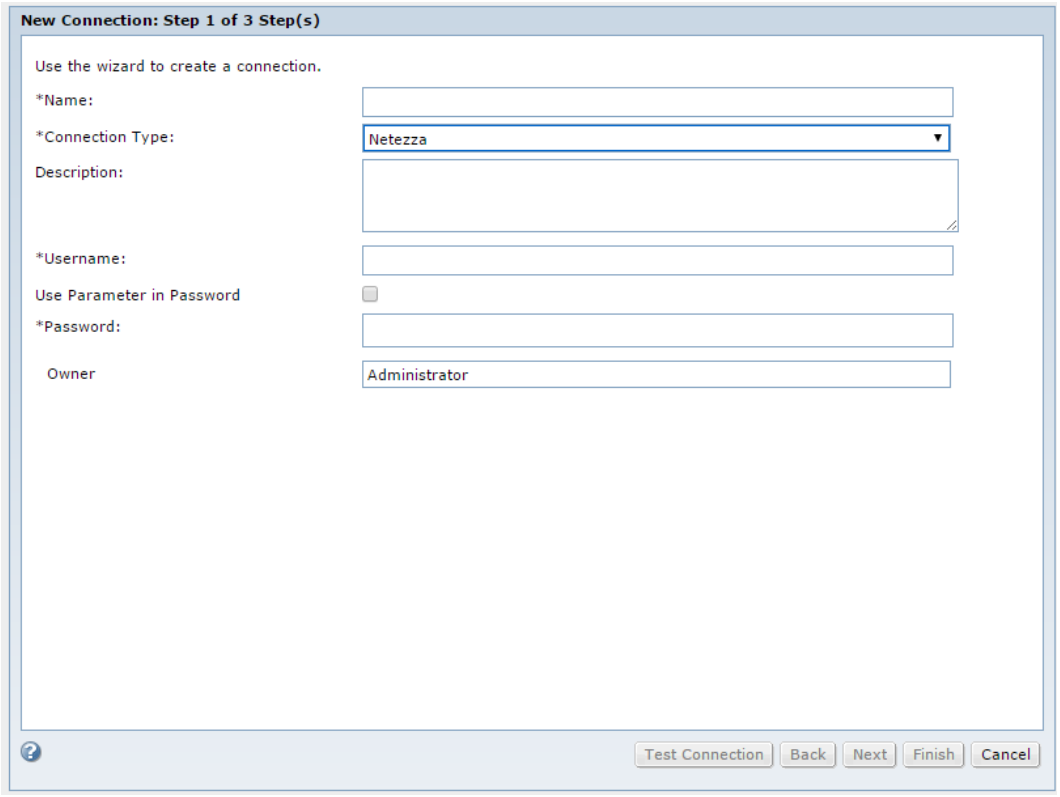
   A tab opens to display the new connection properties.

2. Enter a connection name and optional description.

   The connection name must begin with an alphabetic character. If you enter a connection name that begins with a numeric character, a workflow that you use the connection in might fail.

3. Select the Netezza connection type and enter the user information.

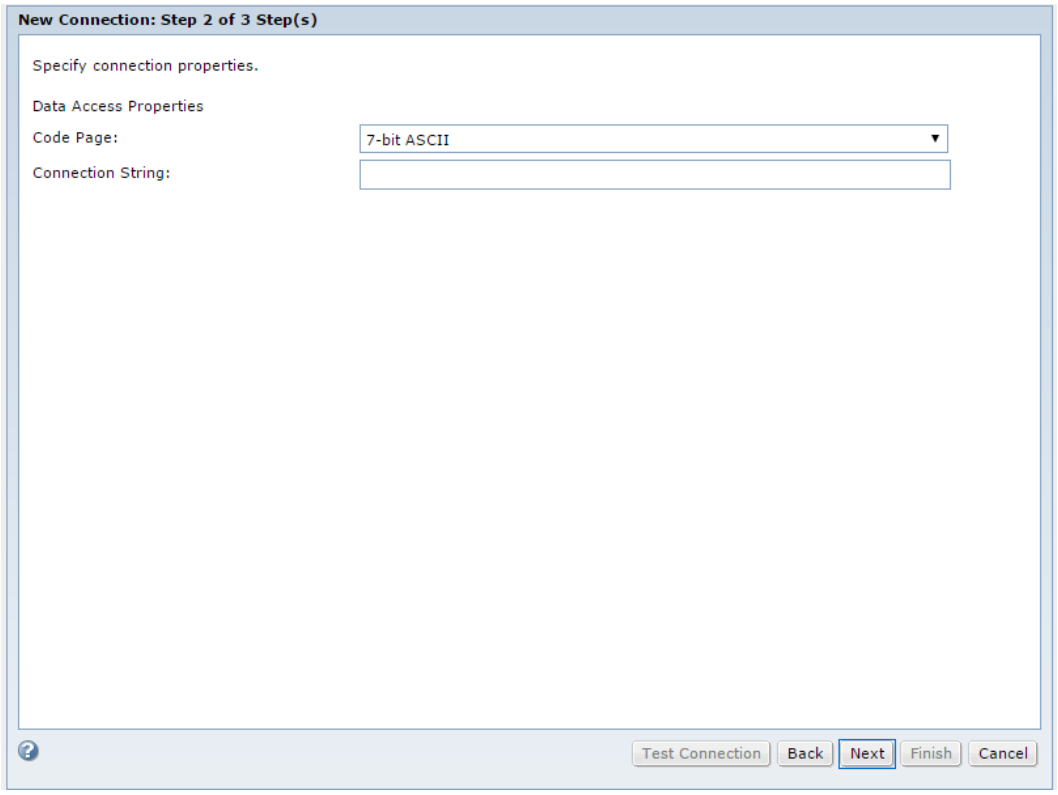
The following image shows the first step of the connection creation wizard:



4. Optional. Click **Change Owner** and select a different user as the connection owner.
5. Click **Next**.
6. Enter the connection properties and click **Next**.

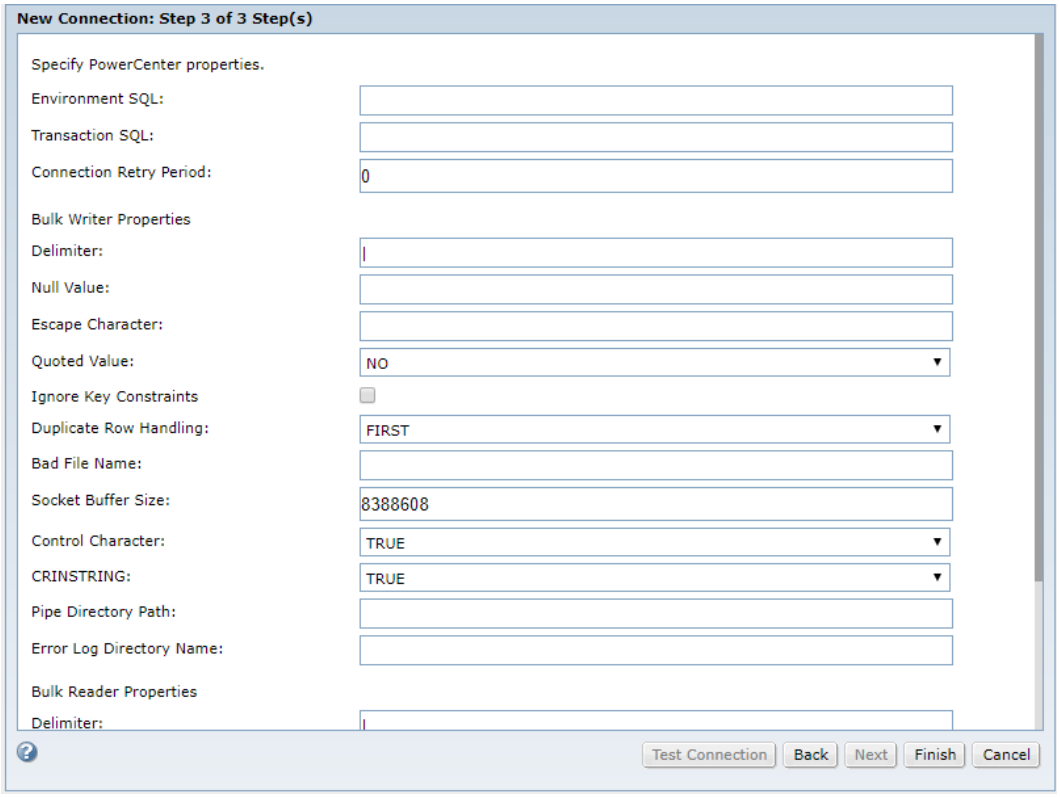
   The connection string must be the name of the connection that you configured in the `odbc.ini` file in the Informatica server.

The following image shows the second step of the connection creation wizard:



7. Enter the PowerCenter properties and click **Next**. Netezza Bulk read and write properties appear starting from some EBFs on 9.7.1 and later versions.

The following image shows the third step of the connection creation wizard:



8. Click **Finish** to save the connection.

   The connection is visible in the **Administrator | Connections** view.

You can use the connection in a data subset or data masking plan.

Author

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